

Classwork: TypeScript Arrays and Loops

100 points + 10 bonus | Predict first, then run in VS Code

Student name

Date

Part 1 - Loop Readiness Check

10 points

```
for (let i = 0; i < 4; i++) {  
  console.log(i);  
}
```

Predict the starting value and all values that print:

What check becomes false and stops the loop?

Why does 4 not print?

Did the real output match your prediction?

Readiness rule

If this trace is confusing, ask for help before moving to array indexes.

Build Typed Arrays

Classwork continued

Part 2 - Build Typed Arrays

15 points

Create both arrays. Then explain what each type annotation allows inside the list.

```
const players: string[] = ["Maya", "Leo", "Zara"];  
const scores: number[] = [78, 91, 84];
```

What does `string[]` tell TypeScript?

What does `number[]` tell TypeScript?

How many variables store all six starting values?

Rewrite both arrays after adding one matching item to each:

Predict what TypeScript reports if `scores` contains the string `high`:

Index and Length Lab

Classwork continued

Part 3 - Index and Length Lab

20 points

```
const colors: string[] = ["red", "green", "blue", "gold"];
```

Record a prediction and actual result for each expression:

colors[0] | colors[1] | colors[3] | colors.length | colors[colors.length - 1]

--

Explain the first index, final index, and why length is different:

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Write one line that changes green to lime:

--

What appears for colors[4], and why?

--

Remember

Length is the number of items. The final valid index is length - 1.

Trace an Array Loop

Classwork continued

Part 4 - Trace an Array Loop

20 points

```
const planets: string[] = ["Mercury", "Venus", "Earth"];

for (let i = 0; i < planets.length; i++) {
  console.log(`Index ${i}: ${planets[i]}`);
}
```

Trace rounds 1, 2, 3, and the final stop check:

What are the two jobs of i?

Why use planets.length instead of typing 3?

What code changes if a fourth planet is added?

AI Training Data Checker

Classwork continued

Part 5 - AI Training Data Checker

25 points

This program flags uncertain labels for a person to inspect. It is a data-review simulation, not an AI model.

```
const labels: string[] = ["cat", "unknown", "dog",  
  "unknown", "bird"];  
let reviewCount: number = 0;  
for (let i = 0; i < labels.length; i++) {  
  const label: string = labels[i];  
  if (label === "unknown") {  
    console.log(`Example ${i + 1} needs human review.`);  
    reviewCount++;  
  } else {  
    console.log(`Example ${i + 1}: ${label}`);  
  }  
}  
console.log(`${reviewCount} examples need human review.`);
```

Predict all output lines, review indexes, and the final count:

Explain the accumulator, $i + 1$, the simulation, and human review:

Human responsibility

A checker can flag uncertainty. A person still decides whether a label is correct.

Repair, Bonus, and Submit

Classwork conclusion

Part 6 - Repair the Boundary Bug

10 points

```
const missions: string[] = ["Map", "Build", "Test"];
for (let i = 0; i <= missions.length; i++) {
  console.log(missions[i]);
}
```

Explain the attempted indexes, unexpected output, and one-character fix:

Bonus - Add Another Review Rule

+10 points

Use seven labels, keep unknown, add a familiar second rule, and trace two rounds.
Explain why the loop stops correctly.

Submit to Microsoft Teams AND Ishwari Raut ma'am.